

Ilker Demirel

PhD candidate at MIT CSAIL

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Interests: LLM agents, self-improvement & continual learning, sample-efficiency, causality, human-AI collaboration

Education

- 2022–now **Massachusetts Institute of Technology**, PhD
Computer Science, GPA – 5.00/5.00
Advisor: David Sontag
Minor: Human Evolutionary Biology (Harvard University)
- 2020 and 2022 **Bilkent University**, BS and MS
Electrical and Electronics Engineering, GPA – 3.90/4.00 (BS), 3.91/4.00 (MS)

Research

- Summer'26 **PhD Research Intern**, Layer Health
◦ Self-improving agentic LLM systems for clinical decision guidance with human experts in the loop
- Summer'25 **PhD Research Intern**, Apple
◦ Evaluation benchmarks and methods for multimodal (audio, image, text) reasoning with LLMs
- Summer'24 **PhD Research Intern**, Microsoft
◦ Scalable and efficient representation learning algorithms for the personal adaptive immune system
- 2022–now **Graduate Student (PhD)**, Massachusetts Institute of Technology
◦ LLMs, representation learning, sample-efficient statistical and causal inference, health AI
- 2022–2024 **PhD Fellow**, Eric and Wendy Schmidt Center, The Broad Institute of MIT and Harvard
- 2020–2022 **Graduate Student (MS)**, Bilkent University
◦ Reinforcement learning, Bayesian optimization, multi-armed bandits
- 2019–2020 **Undergrad Senior Project**, NanoMagnetism Instrument and Bilkent University
◦ Safe online learning algorithms to tune an Atomic Force Microscope
- Summer'19 **Undergrad Intern**, Robotics and Control Laboratory, University of British Columbia
◦ Deep reinforcement learning for point-of-care ultrasound
- Summer'18 **Undergrad Intern**, DataBoss Security and Analytics
◦ ML-based anomaly detection algorithms for cyber-security

Teaching

- Summer'25 **Instructor**, *Causal Inference Workshop*, Paris AI4Health School
- Spring'25 **Teaching Assistant**, 6.7930/HST.956 *ML for Healthcare*, Massachusetts Institute of Technology
- Spring'22 **Teaching Assistant**, EEE485/585 *Statistical Learning and Data Analytics*, Bilkent University
- F'20'21, S'21 **Teaching Assistant**, EEE212 *Microprocessors*, Bilkent University

Publications (*equal contribution)

Preprint

- [P1] LLMs can construct powerful representations and streamline sample-efficient supervised learning [paper]
ID, L. Shi, Z. Hussain, D. Sontag, *arXiv*.
- [P2] Machine learning cross-platform proteomic imputation enables protein quality scoring and replication of epidemiological associations [paper]
L. Li, A. Alaa, Y. Tan, ID, ..., P. Natarajan, Z. Yu, *bioRxiv*.

Conference

- [C11] Uncovering bias mechanisms in observational studies [\[paper\]](#)
ID*, Z. Hussain*, P.D. Bartolomeis, D. Sontag, *ICML 2026*.
- [C10] Prediction-powered causal inferences [\[paper\]](#)
R. Cadei, ID*, P.D. Bartolomeis*, L. Lindorfer, S. Cremer, C. Schmid, F. Locatello, *NeurIPS 2025*.
- [C9] Using LLMs for late multimodal sensor fusion for activity recognition [\[paper\]](#)
ID, K. Thakkar, B. Elizalde, S. Ren, J. Narain, *NeurIPS 2025, Time Series for Health Workshop*.
- [C8] Scalable universal T-cell receptor embeddings from adaptive immune repertoires [\[paper\]](#)
P. Chapfuwa, ID, L. Pisani, J. Zazo, E. Portugaly, H. Zahid, J. Greissl, *ICLR 2025*.
- [C7] Prediction-powered generalization of causal inferences [\[paper\]](#)
ID, A. Alaa, A. Philippakis, D. Sontag. *ICML 2024*.
- [C6] Benchmarking observational studies with experimental data under right-censoring [\[paper\]](#)
ID, E.D. Brouwer, M. Oberst, Z. Hussain*, A. Philippakis, D. Sontag. *AISTATS 2024*.
- [C5] Association of Eicosanoid Metabolites With Cardiac Structure and Function [\[paper\]](#)
A. Roshandelpoor, J. Ma, ID, E. Lau, M. Alotaibi, M.F. Ramirez Fernandez Del Castillo, J. Parekh, A. Pan, N. Owunna, V. Varadarajan, M. Jain, S. Cheng, J.A.C.A. Lima, J.E. Ho, *AHA Circulation, 2023*.
- [C4] Falsification of internal and external validity in observational studies via conditional moment restrictions [\[paper\]](#)
Z. Hussain*, M.C. Shih*, M. Oberst, ID, D. Sontag. *AISTATS 2023*.
- [C3] Escada: Efficient safety and context aware dose allocation for precision medicine [\[paper\]](#)
ID, A.A. Celik, C. Tekin. *NeurIPS 2022*.
- [C2] Combinatorial gaussian process bandits with probabilistically triggered arms [\[paper\]](#)
ID, C. Tekin. *AISTATS 2021*.
- [C1] Accuracy limits of distance estimation in visible light systems with RGB LEDs [\[paper\]](#)
ID, S. Gezici. *IEEE PIMRC 2019*.

Journal

- [J2] Federated multi-armed bandits under byzantine attacks [\[paper\]](#)
A. Saday, ID, Y. Yildirim, S. Gezici. *IEEE Transactions on Artificial Intelligence, 2025*.
- [J1] Distance and position estimation in visible light systems with RGB LEDs [\[paper\]](#)
ID, S. Gezici. *DSP Elsevier, 2022*.

Invited Talks

- Apr-May 2026 **MIT IMES Seminar, ShahLab at Stanford, Stripe**, LLM rubric representation learning
- Jul 2025 **Paris AI4Health School**, Delivered a workshop on causal inference for precision medicine
- Feb 2025 **Harvard Data Science Initiative Causal Seminar**, Served as the discussant
- Oct 2024 **MIT LIDS Postdoc Seminar**, Prediction-powered causal inference
- March 2022 **Uludag University School of Medicine**, Personalized type-1 diabetes treatment

Honors and Awards

- 2022–2024 **PhD Fellowship**, Eric and Wendy Schmidt Center at the Broad Institute of MIT and Harvard
- 2020–2022 **5G and Beyond Graduate Fellowship**, Vodafone and Bilkent University
- 2015–2020 **Comprehensive Undergraduate Scholarship**, Bilkent University
- 2015–2020 **Academic Excellence Scholarship**, Turkey Higher Education Student Loan and Housing Board

Reviewing

NeurIPS 2025, ICML 2026 – Top Reviewer
JMLR, NeurIPS, ICML, ICLR, AISTATS, AAAI, ML4H, IEEE